



JABchem



Not to be shared without the copyright holder's permission

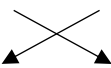
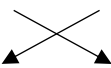
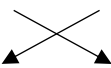
Past Papers Nat 5 Chemistry

2025 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	89+	71%	45.3%
B	75+	60%	19.9%
C	62+	50%	14.5%
D	48+	38%	11.7%
No award	<48	<38%	8.6%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.8 /25	46.5 /75	18.1 /25

2025 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning								
1	A	90	☒ A Elements in group 0 usually have 8 electron in their outer shell (except helium with 2) ☒ B Elements in group 1 have 1 electron in their outer shell ☒ C Elements in group 2 have 2 electrons in their outer shell ☒ D Elements in group 7 have 7 electrons in their outer shell								
2	C	80	☒ A atomic number from nuclide notation is 26 $\therefore$ atom has 26 protons ☒ B atomic number from nuclide notation is 26 $\therefore$ atom has 26 protons ☒ C atom has 26 protons as atomic no = 26 and 26 electrons as no. of protons = electrons ☒ D atoms have no charge overall so number of protons = number of electrons = 26								
3	D	72	☒ A Phosphorus chloride has formula PCl_3 $\therefore$ PCl_3 molecule contains 4 atoms per molecule. ☒ B Magnesium Chloride MgCl_2 is ionic & contains no molecules. It contains 3 ions not atoms. ☒ C Sodium fluoride NaF is ionic & contains no molecules. It contains 2 ions not atoms. ☒ D Carbon dioxide has formula CO_2 $\therefore$ CO_2 molecule contains 3 atoms per molecule.								
4	B	43	☒ A Covalent molecular bonding is found in elements e.g. H_2 , N_2 , O_2 , F_2 , Cl_2 ☒ B Ionic lattices are never found in elements but found in compounds (metal joined to non-metal) ☒ C Metallic Lattice structure is found in metallic elements e.g. K, Na, Fe, Mg, Al ☒ D Covalent network is found in non-metal elements with a high melting point e.g. C, B, Si								
5	B	63	☒ A silicon dioxide SiO_2 is a covalent compound and does not conduct in any state ☒ B lithium bromide LiBr is ionic, melts at 550°C and conducts in the liquid state at 600°C ☒ C ammonia NH_3 is a covalent compound and does not conduct in any state ☒ D barium chloride BaCl_2 is ionic, melts at 961°C but will not conduct at 600°C as it is a solid								
6	A	73	☒ A calcium oxide is a soluble metal oxide which dissolves in water to form an alkali ($\text{pH} > 7$) ☒ B nitrogen dioxide is a non-metal oxide and dissolves in water to form an acid ($\text{pH} < 7$) ☒ C sulfur dioxide is a non-metal oxide and dissolves in water to form an acid ($\text{pH} < 7$) ☒ D nickel oxide is a insoluble metal oxide but does not change the pH of water								
7	D	75	☒ A adding water decreases the concentration of the acid and decreases reaction rate ☒ B adding water decreases the concentration of the H^+ ions in the acid ☒ C adding water does not neutralise the acid as water must be formed in a neutralisation ☒ D adding water decreases the concentration of H^+ ions which increases pH up to $\text{pH} = 7$								
8	B	74	$\text{gfm} (\text{NH}_4)_3\text{PO}_4 = 149\text{g}$ $\% \text{P} = \frac{\text{mass of N}}{\text{gfm}} \times 100 = \frac{3 \times 14}{149} \times 100 = \frac{42}{149} \times 100 = 28.2\%$								
9	A	64	Mn ion must have the same number of charge as the dichromate $\text{Cr}_2\text{O}_7^{2-}$ ion but the opposite charge to make formula neutral overall $\therefore$ Mn has 2+ charge <table border="1"> <thead> <tr> <th>Write down Symbols and valency below</th><th>Cross-Over arrows to work out formula</th><th>Work out chemical formula (Cancel down if necessary)</th><th>Ionic Formula</th></tr> </thead> <tbody> <tr> <td> $\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$ 2 2 </td><td> $\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$  2 2 </td><td> $\text{Mn Cr}_2\text{O}_7$ </td><td> $\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$ </td></tr> </tbody> </table>	Write down Symbols and valency below	Cross-Over arrows to work out formula	Work out chemical formula (Cancel down if necessary)	Ionic Formula	$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$ 2 2	$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$  2 2	$\text{Mn Cr}_2\text{O}_7$	$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$
Write down Symbols and valency below	Cross-Over arrows to work out formula	Work out chemical formula (Cancel down if necessary)	Ionic Formula								
$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$ 2 2	$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$  2 2	$\text{Mn Cr}_2\text{O}_7$	$\text{Mn}^{2+} \text{Cr}_2\text{O}_7^{2-}$								
10	C	75	☒ A propane C_3H_8 has more than 2 hydrogen atoms for every carbon ☒ B propan-1-ol $\text{C}_3\text{H}_7\text{OH}$ contains the hydroxyl $-\text{OH}$ group and is soluble in water ☒ C cyclopropane C_3H_6 is saturated, insoluble in water and has 2 hydrogens for every carbon ☒ D propene C_3H_6 is unsaturated as it contains a $\text{C}=\text{C}$ double bond								

11	D	82	Structure														
			Formula	C ₃ H ₄	C ₄ H ₆	C ₅ H ₈											
			Relationship	If n=3 then 2n-2=4	If n=4 then 2n-2=6	If n=5 then 2n-2=8											
			General Formula	C _n H _{2n-2}	C _n H _{2n-2}	C _n H _{2n-2}											
12	D	47	☒ A gfm = 16 ∴ formula of hydrocarbon = CH ₄ ∴ CH ₄ is methane from the alkane homologous series ☒ B gfm = 44 ∴ formula of hydrocarbon = C ₃ H ₈ ∴ C ₃ H ₈ is propane from the alkane homologous series ☒ C gfm = 58 ∴ formula of hydrocarbon = C ₄ H ₁₀ ∴ C ₄ H ₁₀ is butane from the alkane homologous series ☒ D gfm = 70 ∴ formula of hydrocarbon = C ₅ H ₁₀ ∴ C ₅ H ₁₀ is not an alkane (either butene or cyclobutene)														
13	C	64	☒ A structure contains four carbons so must start with the prefix <i>but-</i> not start with <i>prop-</i> ☒ B Structure contains the carboxyl -COOH group so cannot be an alcohol ☒ C structure contain carboxyl -COOH group and four carbons ∴ name is butanoic acid ☒ D Structure contains the carboxyl -COOH group so cannot be an alcohol														
14	A	87	CH ₃ CHC(OH)C(CH ₃) ₃ has molecular formula C ₇ H ₁₄ O ☒ A structure has molecular formula C ₇ H ₁₄ O ☒ B structure has molecular formula C ₈ H ₁₈ O ☒ C structure has molecular formula C ₈ H ₁₆ O ☒ D structure has molecular formula C ₇ H ₁₄ O														
15	B	61	☒ A two products formed are butan-1-ol and butan-2-ol ☒ B one product formed called butan-1-ol ☒ C two products formed are hexan-1-ol and hexan-2-ol ☒ D two products formed are hexan-2-ol and hexan-3-ol														
16	C	89	$\begin{array}{ccc} 5g & \longleftrightarrow & 150kJ \\ 60g & \longleftrightarrow & 150kJ \times \frac{60}{5} \\ & & = 1800kJ \end{array}$														
17	B	64	☒ A The positive ions in metallic bonding are a combination of the nucleus and the inner shells ☒ B There are no negative ions involved in metallic bonding ☒ C The delocalised electrons in metallic bonding are the electrons ion the outer shell ☒ D Positive ions and delocalised electrons are held together by electrostatic attraction														
18	A	88	☒ A metal X is aluminium as it gives a voltage lower than magnesium (2.7V) but higher that zinc (1.1V) due to aluminium being between magnesium and zinc in the ECS. ☒ B metal X cannot be sodium as it is above magnesium in ECS and would give voltage less than 2.7V ☒ C metal X cannot be nickel as it is below iron in ECS and would give voltage less than 0.9V ☒ D metal X cannot be copper as it is below iron in ECS and would give voltage less than 0.9V														
19	D	47	☒ A reaction is redox as oxidation (Zn → Zn ²⁺ + 2e ⁻) and reduction (2H ⁺ + 2e ⁻ → H ₂) are taking place ☒ B reaction is redox as oxidation (Cu → Cu ²⁺ + 2e ⁻) and reduction (2Cl ⁻ + 2e ⁻ → Cl ₂) are taking place ☒ C reaction is redox as oxidation (2Fe ²⁺ → 2Fe ³⁺ + 2e ⁻) and reduction (2Br ⁻ + 2e ⁻ → Br ₂) are taking place ☒ D reaction is precipitation as an insoluble solid is formed														
20	D	82	☒ A water is a product of the Ostwald Process ☒ B oxygen is a reactant in the Ostwald Process ☒ C ammonia is a reactant in the Ostwald Process ☒ D platinum is the catalyst used in the Ostwald Process														
21	A	70	<table><tr><td>Particle</td><td>Proton</td><td>Neutron</td><td>Electron</td><td>Alpha</td><td>Beta</td></tr><tr><td>Symbol</td><td>1_1p</td><td>1_0n</td><td>${}^0_{-1}e$</td><td>4_2He</td><td>${}^0_{-1}e$</td></tr></table>	Particle	Proton	Neutron	Electron	Alpha	Beta	Symbol	1_1p	1_0n	${}^0_{-1}e$	4_2He	${}^0_{-1}e$		
Particle	Proton	Neutron	Electron	Alpha	Beta												
Symbol	1_1p	1_0n	${}^0_{-1}e$	4_2He	${}^0_{-1}e$												
22	C	79	☒ A lithium sulfate is soluble and cannot be made by precipitation ☒ B sodium sulfate is soluble and cannot be made by precipitation ☒ C barium sulfate is insoluble and made by precipitation (collected by filtration) ☒ D magnesium sulfate is soluble and cannot be made by precipitation														

23	C	82	☒ A The volume displayed is between 9cm ³ and 10cm ³ so cannot be 8.8cm ³ ☒ B The bottom of the meniscus is used to measure volume (9.1cm ³ uses top of meniscus) ☒ C The bottom of the meniscus is used to measure volume and volume is 9.2cm ³ ☒ D The volume displayed is between 9cm ³ and 10cm ³ so cannot be 10.8cm ³
24	D	52	Titration 1 volume is ignored as it is the rough titration and not designed to be accurate. Titration 3 is not used in the calculation of average titre as it is not concordant as it is more the 0.2cm³ away from titrations 2 and 4. $\text{average titre} = \frac{10.1 + 10.3}{2} = \frac{20.4}{2} = 10.2\text{cm}^3$
25	B	66	☒ A A burette is not as accurate as a pipette for measuring exactly 10cm ³ ☒ B A pipette is the best piece of apparatus to measure exactly 10cm ³ of solution. ☒ C A conical flask will not measure 10cm ³ of solution accurately ☒ D A measuring cylinder will not measure 10cm ³ of solution accurately

2025 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning								
1a(i)	Hydrogen Burns with a pop	<table><tr><td>Gas</td><td>Hydrogen</td><td>Oxygen</td><td>Carbon Dioxide</td></tr><tr><td>Test</td><td>Burns with a pop</td><td>Relights a glowing splint</td><td>Turns limewater cloudy</td></tr></table>	Gas	Hydrogen	Oxygen	Carbon Dioxide	Test	Burns with a pop	Relights a glowing splint	Turns limewater cloudy
Gas	Hydrogen	Oxygen	Carbon Dioxide							
Test	Burns with a pop	Relights a glowing splint	Turns limewater cloudy							
1a(ii)	Gas escapes from conical flask as not stopper	The conical flask lacks a stopper at the mouth of the flask. Gas will escape from the mouth of the flask rather than be forced along the delivery tube at collect over water in the measuring cylinder.								
1a(iii)	evaporation/boiling	To remove the water (solvent) from a solution then heat will speed up the evaporation of the solvent leaving the solute behind.								
1b(i)	1.72	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{86 - 0}{50 - 0} = 1.72 \text{ cm}^3 \text{ s}^{-1}$								
1b(ii)	Curve drawn with:	<table><tr><td>Less steep gradient at start</td><td>Same final volume of gas (96cm³)</td></tr></table>	Less steep gradient at start	Same final volume of gas (96cm ³)						
Less steep gradient at start	Same final volume of gas (96cm ³)									
1b(iii)	One answer from:	<table><tr><td>If zinc in excess: Final volume of gas would be half = 48cm³</td><td>If acid in excess: Same volume of gas released = 96cm³</td></tr></table>	If zinc in excess: Final volume of gas would be half = 48cm ³	If acid in excess: Same volume of gas released = 96cm ³						
If zinc in excess: Final volume of gas would be half = 48cm ³	If acid in excess: Same volume of gas released = 96cm ³									
2a	Diagram showing:									
2b	Sulfur or other group 6 element	Elements in the same group of the Periodic Table have similar chemical properties as they have the same number of electrons in their outer shells.								
2c	Increase in temperature decreases the solubility	Problem Solving: interpreting information from a line graph. Cause and Effect means the changing temperature causes the changing solubility								
2d(i)	angular	<table><tr><td></td><td></td><td></td><td></td></tr><tr><td>linear</td><td>angular</td><td>trigonal pyramidal</td><td>tetrahedral</td></tr></table>					linear	angular	trigonal pyramidal	tetrahedral
linear	angular	trigonal pyramidal	tetrahedral							
2d(ii)	positive electrons	A covalent bond forms when two positive nuclei are held together by their common attraction for a shared pair of electrons .								
3a(i)		<table><tr><td></td><td></td><td></td></tr><tr><td>Monomer</td><td>Polymer</td><td>Repeating Unit</td></tr></table>				Monomer	Polymer	Repeating Unit		
Monomer	Polymer	Repeating Unit								
3a(ii)	Diatomic	A molecule containing only two atoms is called diatomic. They can be elements e.g. O ₂ or compounds e.g. HCl								

3b(i)	Bar Graph showing:	1 mark	1 mark		1 mark	1 mark																									
		For appropriate format: bars (not 5 points)	The 'melting point' axis of the graph has a suitable scale. For the graph paper provided within the question paper, the selection of a suitable scale will result in a graph (plotted bars) that occupies greater than half of the width and half of the height of the graph paper.		The axes of the graph have suitable labels and units.	All bars are plotted accurately (within a half box tolerance).																									
3b(ii)	PVC would melt at temperature above 82°C	Water boils at 100°C and PVC is the only plastic with a melting point below 100°C and would melt when the temperature of the water rose above 82°C																													
4a	carboxyl group		—NH_2	—O—H	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—C—OH} \end{array}$																										
			amino group	hydroxyl group	carboxyl group																										
4b(i)	hydrogenation	Addition reactions add a molecule across a C=C double bond - addition of hydrogen is also known as hydrogenation - addition of water is also known as hydration																													
4b(ii)	Speeds up reaction but not used up	A catalyst speeds up a chemical reaction but is not used up in the reaction																													
4c	Diagram showing:																														
4d	Ibuprofen	Ibuprofen is the only medicine that has a use in the treatment of pain and the question states ibuprofen works by blocking the body's response to pain.																													
5a	burns to release energy	Fuels are chemical compounds which release energy when they are burned. They are exothermic reactions.																													
5b(i)	Structure showing:																														
5b(ii)	Iso-butene would decolourise bromine solution	Iso-butene contains a C=C double bond which would decolourise bromine solution quickly. Iso-butane has not C=C double bonds so would not decolourise bromine solution,																													
5c(i)	more branches lower flashpoint	<table><tr><td>Name</td><td>Number of branches</td><td>Flashpoint (°C)</td></tr><tr><td>octane</td><td>0</td><td>13</td></tr><tr><td>2-methylheptane</td><td>1</td><td>4</td></tr><tr><td>2,2-dimethylhexane</td><td>2</td><td>-3</td></tr><tr><td>2,2,4-trimethylpentane</td><td>3</td><td>-12</td></tr></table>					Name	Number of branches	Flashpoint (°C)	octane	0	13	2-methylheptane	1	4	2,2-dimethylhexane	2	-3	2,2,4-trimethylpentane	3	-12										
		Name	Number of branches	Flashpoint (°C)																											
octane	0	13																													
2-methylheptane	1	4																													
2,2-dimethylhexane	2	-3																													
2,2,4-trimethylpentane	3	-12																													
5c(ii)	29	<table><tr><td>Compound</td><td>Hexane</td><td>Heptane</td><td>Octane</td><td>Nonane</td></tr><tr><td>Formula</td><td>C₆H₁₄</td><td>C₇H₁₆</td><td>C₈H₁₈</td><td>C₉H₂₀</td></tr><tr><td>Flash Point (°C)</td><td>-22</td><td>-4</td><td>13</td><td>-</td></tr><tr><td>Difference (°C)</td><td></td><td>18</td><td>17</td><td>(16)</td></tr><tr><td>Prediction (°C)</td><td>-</td><td>-</td><td>-</td><td>29</td></tr></table>					Compound	Hexane	Heptane	Octane	Nonane	Formula	C ₆ H ₁₄	C ₇ H ₁₆	C ₈ H ₁₈	C ₉ H ₂₀	Flash Point (°C)	-22	-4	13	-	Difference (°C)		18	17	(16)	Prediction (°C)	-	-	-	29
		Compound	Hexane	Heptane	Octane	Nonane																									
		Formula	C ₆ H ₁₄	C ₇ H ₁₆	C ₈ H ₁₈	C ₉ H ₂₀																									
		Flash Point (°C)	-22	-4	13	-																									
		Difference (°C)		18	17	(16)																									
Prediction (°C)	-	-	-	29																											
5d(i)	Contain oxygen needed for combustion	Oxygen is needed for combustion to happen. If the chemical being burned does not contain oxygen then all the oxygen will need to come from air.																													

5d(ii)	Answer to include:	1 mark ethanol has stronger forces of attraction 1 mark forces between molecules/intermolecular attractions mentioned																
6	Open Question:	<table><tr><td colspan="2">3 mark answer</td><td>2 mark answer</td><td>1 mark answer</td></tr><tr><td colspan="2">Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer		2 mark answer	1 mark answer	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.								
	3 mark answer		2 mark answer	1 mark answer														
Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.															
<table><tr><td colspan="4">Marks Awarded</td></tr><tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr><tr><td>11%</td><td>27%</td><td>52%</td><td>10%</td></tr></table>		Marks Awarded				3 marks	2 marks	1 mark	0 marks	11%	27%	52%	10%					
Marks Awarded																		
3 marks	2 marks	1 mark	0 marks															
11%	27%	52%	10%															
7a	nucleus	Radioactive decay occurs when a nucleus has an unstable ratio of protons to neutrons. Alpha or beta particles are emitted to change the proton to neutron ratio back towards a more stable ratio.																
7b(i)	Sodium, hydrogen and oxygen	<table><tr><td>Ending</td><td>Meaning</td></tr><tr><td>-ide</td><td>2 named elements in compound</td></tr><tr><td>-ate</td><td>2 named elements in compound plus oxygen</td></tr><tr><td>-ite</td><td>2 named elements in compound plus oxygen</td></tr></table>	Ending	Meaning	-ide	2 named elements in compound	-ate	2 named elements in compound plus oxygen	-ite	2 named elements in compound plus oxygen								
Ending	Meaning																	
-ide	2 named elements in compound																	
-ate	2 named elements in compound plus oxygen																	
-ite	2 named elements in compound plus oxygen																	
7b(ii)	Same	The half-life of a radioisotopes is decided by the stability of the proton tio neutron ratio. State of matter (solid/liquid/gas/solution) or being an element or compound does not change the half-life.																
7c(i)	Xenon or Xe	${}_{53}^{131}\text{I} \longrightarrow {}_{54}^{131}\text{Xe} + {}_{-1}^0\text{e}$																
7c(ii)	8	<table><tr><td>Mass (g)</td><td>No. of half-lives</td></tr><tr><td>40</td><td>0</td></tr><tr><td>20</td><td>1</td></tr><tr><td>10</td><td>2</td></tr><tr><td>5</td><td>3</td></tr><tr><td>2.5</td><td>4</td></tr></table> 4 half-lives = 32 days ∴ 1 half-life = 8 days	Mass (g)	No. of half-lives	40	0	20	1	10	2	5	3	2.5	4				
Mass (g)	No. of half-lives																	
40	0																	
20	1																	
10	2																	
5	3																	
2.5	4																	
7d(i)	Scale as shown: (1 mark)																	
7d(ii)	label as shown: (1 mark)																	
8a	mass	Problem Solving: Extraction of information from passage																
8b	${}_{6}^{12}\text{C}$	The atomic number of carbon is always 6. All carbon atoms have 6 protons. Atomic number is the bottom number in front of the element symbol in nuclide notation The mass number of this carbon isotope must be 12. The mass number is the top number in front of nuclide notation.																
8c	18g	1 mol H ₂ O = (2x1)+(1x16) = 2+16 = 18g																
8d	Electrons have almost zero mass	<table><tr><td>Particle</td><td>Location</td><td>Charge</td><td>Mass</td></tr><tr><td>proton</td><td>inside nucleus</td><td>+1</td><td>1 amu</td></tr><tr><td>neutron</td><td>inside nucleus</td><td>0</td><td>1 amu</td></tr><tr><td>electron</td><td>outside nucleus</td><td>-1</td><td>almost 0</td></tr></table>	Particle	Location	Charge	Mass	proton	inside nucleus	+1	1 amu	neutron	inside nucleus	0	1 amu	electron	outside nucleus	-1	almost 0
Particle	Location	Charge	Mass															
proton	inside nucleus	+1	1 amu															
neutron	inside nucleus	0	1 amu															
electron	outside nucleus	-1	almost 0															
8e	5 5L	CH ₄ has five atoms per molecule (1 carbon and 4 hydrogen atoms) 5 atoms per molecule means 5L atoms in one mole																
9a(i)	isomers	Same molecular formula different structural formula																

9a(ii)	One name and structure from:	propan-1-ol $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{OH} \\ & & \\ \text{H} & \text{H} & \text{H} \end{array}$ or propan-2-ol $\begin{array}{c} \text{H} & \text{H} & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ & & \\ \text{H} & \text{OH} & \text{H} \end{array}$												
9b	$\text{C}_3\text{H}_7\text{OH} + 4\frac{1}{2}\text{O}_2$ ↓ $3\text{CO}_2 + 4\text{H}_2\text{O}$	$\text{C}_3\text{H}_7\text{OH} + 4\frac{1}{2}\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$ or any multiple including: $2\text{C}_3\text{H}_7\text{OH} + 9\text{O}_2 \rightarrow 6\text{CO}_2 + 8\text{H}_2\text{O}$												
9c(i)	Reduce heat loss to surroundings	Draughts remove heat energy from the experiment to the surrounding reducing. This means the temperature rise in the water is not as high as it could be.												
9c(ii)A	21	Average temperature = $\frac{23+20+20}{3} = \frac{63}{3} = 21^\circ\text{C}$												
9c(ii)B	8.78	heat energy = specific heat capacity × mass × change in Temperature $E_h = C \times m \times \Delta T$ $E_h = 4.18 \times 0.1 \times 21$ $E_h = 8.78 \text{ kJ}$												
10a(i)	solvent	<table border="1"><thead><tr><th>Term</th><th>Meaning</th></tr></thead><tbody><tr><td>Solute</td><td>The solid dissolved in the solvent</td></tr><tr><td>Solvent</td><td>The liquid that does the dissolving</td></tr><tr><td>Solution</td><td>The mixture formed when a solute dissolves in a solvent</td></tr></tbody></table>	Term	Meaning	Solute	The solid dissolved in the solvent	Solvent	The liquid that does the dissolving	Solution	The mixture formed when a solute dissolves in a solvent				
Term	Meaning													
Solute	The solid dissolved in the solvent													
Solvent	The liquid that does the dissolving													
Solution	The mixture formed when a solute dissolves in a solvent													
10a(ii)A	ethanol	<table border="1"><tbody><tr><td>Monomer</td><td>ethene</td><td>propene</td><td>chloroethene</td><td>phenylethene</td><td>ethanol</td></tr><tr><td>Polymer</td><td>poly(ethene)</td><td>poly(propene)</td><td>poly(chloroethene)</td><td>poly(phenylethene)</td><td>poly(ethanol)</td></tr></tbody></table>	Monomer	ethene	propene	chloroethene	phenylethene	ethanol	Polymer	poly(ethene)	poly(propene)	poly(chloroethene)	poly(phenylethene)	poly(ethanol)
Monomer	ethene	propene	chloroethene	phenylethene	ethanol									
Polymer	poly(ethene)	poly(propene)	poly(chloroethene)	poly(phenylethene)	poly(ethanol)									
10a(ii)B	No C=C double bond in ethanol	Monomers in to process of addition polymerisation require a C=C double bond for the mechanism of polymerisation to work. Ethanol does not have C=C bond in its structure.												
10b(i)	Titration	Titration is the name of technique where burettes are used to calculate exact volumes of solutions with accurately known concentrations to work out quantities of other solutions using balanced equations.												
10b(ii)	To show HCl is completely neutralised	The indicator is used to show the moment of neutralisation when all the acid and base have reacted completely.												
10b(iii)	0.152	number of moles = volume × concentration = 0.0190 litres × 0.1 mol l ⁻¹ = 0.00190 mol Na_2CO_3 1mol 0.00190mol $:$: : concentration = $\frac{\text{no. of moles}}{\text{volume}}$ HCl 2mol 0.00380mol $= \frac{0.00380 \text{ mol}}{0.025 \text{ litres}} = 0.152 \text{ mol l}^{-1}$												
11a	lithium nitrate	METAL OXIDE + ACID → SALT + WATER lithium oxide + nitric acid → lithium nitrate + water $\text{Li}_2\text{O} + 2\text{HNO}_3 \rightarrow 2\text{LiNO}_3 + \text{H}_2\text{O}$												
11b	3.75g	gfm $\text{Li}_2\text{O} = (2 \times 7) + (1 \times 16) = 14 + 16 = 30\text{g}$ no. of moles = volume × concentration = 0.25 litres × 0.5 mol l ⁻¹ = 0.125mol mass = no. of moles × gfm = 0.125 × 30 = 3.75g												

11c	copper carbonate	ACID + METAL CARBONATE → SALT + WATER + CARBON DIOXIDE sulfuric acid + copper carbonate → copper sulfate + water + carbon dioxide $\text{H}_2\text{SO}_4 + \text{CuCO}_3 \rightarrow \text{CuSO}_4 + \text{H}_2\text{O} + \text{CO}_2$														
11d(i)	exothermic	<table><tr><th>Term</th><th>Meaning</th></tr><tr><td>exothermic</td><td>Reaction or process which releases heat energy</td></tr><tr><td>endothermic</td><td>Reaction or process which takes in heat energy</td></tr></table>			Term	Meaning	exothermic	Reaction or process which releases heat energy	endothermic	Reaction or process which takes in heat energy						
Term	Meaning															
exothermic	Reaction or process which releases heat energy															
endothermic	Reaction or process which takes in heat energy															
11d(ii)A	-100	Enthalpy change = $50 - 150 = -100 \text{ kJ mol}^{-1}$ Negative sign is necessary as the curve finishes below where it starts														
11d(ii)B	<table><tr><td>decrease</td></tr><tr><td>no effect</td></tr></table>	decrease	no effect	The activation energy E_a decreases as the dotted line does not climb as high as the full line when a catalyst is added. The enthalpy change ΔH does not change when a catalyst is added as the starting and finishing positions do not change.												
decrease																
no effect																
12	Open Question: <table><tr><th colspan="4">Marks Awarded</th></tr><tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr><tr><td>18%</td><td>25%</td><td>33%</td><td>23%</td></tr></table>	Marks Awarded				3 marks	2 marks	1 mark	0 marks	18%	25%	33%	23%	3 mark answer Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has been provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
Marks Awarded																
3 marks	2 marks	1 mark	0 marks													
18%	25%	33%	23%													